In [ ]:
```
# dictionary
CGPA ={1:8.9,2:5.6,4:6.7,7:9.1,8:5.3}
print("dictionary  -->",(CGPA))
print("total elements -->", len(CGPA))
print(" type of objects -->",type(CGPA))
print("\n CGPA of 1-->",CGPA[1])
```

```
dictionary  --> {1: 8.9, 2: 5.6, 4: 6.7, 7: 9.1, 8: 5.3}
total elements --> 5
 type of objects --> <class 'dict'>

 CGPA of 1--> 8.9
```

In [ ]:
```
for k in CGPA:
 print("CGPA of",k,"-->",CGPA[k])
```

```
CGPA of 1 --> 8.9
CGPA of 2 --> 5.6
CGPA of 4 --> 6.7
CGPA of 7 --> 9.1
CGPA of 8 --> 5.3
```

In [ ]:
```
for k in CGPA:
 print("CGPA of-->\n",CGPA[k])
```

```
CGPA of-->,k,
 8.9
CGPA of-->,k,
 5.6
CGPA of-->,k,
 6.7
CGPA of-->,k,
 9.1
CGPA of-->,k,
 5.3
```

In [ ]:
```
CGPA[3]=8.6
print("\n after adding(3)-->3",CGPA)
```

```
 after adding(3)-->3 {1: 8.9, 2: 5.6, 4: 6.7, 7: 9.1, 8: 5.3, 3: 8.6}
```

In [ ]:
```
s=0
for i in CGPA:
  s=s=CGPA[i]
print("\n sum-->",s)
print("average-->",s/len(CGPA))

print("average-->",s//len(CGPA))
```

```
 sum--> 8.6
average--> 1.4333333333333333
average--> 1.0
```

In [ ]:
```
CGPA ={1:8.9,2:5.6,4:6.7,7:9.1,8:5.3}
print("\n sum-->",sum(CGPA.values()))
print("average-->",sum(CGPA.values())//len(CGPA))
```

```
 sum--> 35.6
average--> 7.0
```

In [ ]:
```
CGPA ={1:8.9,2:5.6,4:6.7,7:9.1,8:5.3}
print("dictionary -->",CGPA)
print("IS key 2 present-->",2 in CGPA.keys())
print("is key 9 present-->",9 in CGPA.keys())
```

```
dictionary --> {1: 8.9, 2: 5.6, 4: 6.7, 7: 9.1, 8: 5.3}
IS key 2 present--> True
is key 9 present--> False
```

In [ ]:
```
keys=['movie','hero','year']
values=['kantara','rishab',2022]
print("keys-->",keys)
print("values-->",values)

movie_dict={}
for i in range(len(keys)):
 movie_dict[keys[i]]=values[i]
print(movie_dict)
```

```
keys--> ['movie', 'hero', 'year']
values--> ['kantara', 'rishab', 2022]
{'movie': 'kantara', 'hero': 'rishab', 'year': 2022}
```

In [ ]:
```
dict1={'name':'john',10:[2,4,3]}
dict2=dict1
print(dict1)
print(dict2)
print()
print("after doing change")
print()
dict1.popitem()
print(dict1)
print(dict2)
```

```
{'name': 'john', 10: [2, 4, 3]}
{'name': 'john', 10: [2, 4, 3]}

after doing change

{'name': 'john'}
{'name': 'john'}
```

In [ ]:
```
dict1={'name':'john',10:[2,4,3]}
dict2=dict1.copy()
print(dict1)
print(dict2)
print()
print("after doing change")
print()
dict1.popitem()
print(dict1)
print(dict2)
print("change in dict1 will not reflect in dict1")
```

```
{'name': 'john', 10: [2, 4, 3]}
{'name': 'john', 10: [2, 4, 3]}

after doing change

{'name': 'john'}
{'name': 'john', 10: [2, 4, 3]}
change in dict1 will not reflect in dict1
```

In [18]:

```
crops={"rice":120,"wheat":85,"maize":60}
print("dictionary -->",crops)
print("total no of elements-->",len(crops))
for k in crops:
 print("yield of ","wheat","-->",crops["wheat"])
```

```
dictionary --> {'rice': 120, 'wheat': 85, 'maize': 60}
total no of elements--> 3
yield of  wheat --> 85
yield of  wheat --> 85
yield of  wheat --> 85
```

In [40]:

```
crops["pulse"]=45
print("\n after adding(pulse)-->",crops)
crops["rice"]=130
print("\n after updating",crops)
print("dictionary-->",crops)
print("crops-->",crops.keys())
print("yield-->",crops.values())
print(" is sugarcane present-->","sugarcane"in crops.keys())
crops["wheat"]=crops["wheat"]+10
print("\n sum-->",sum(crops.values()))
```

```
 after adding(pulse)--> {'rice': 130, 'wheat': 175, 'maize': 60, 'pulse': 45}

 after updating {'rice': 130, 'wheat': 175, 'maize': 60, 'pulse': 45}
dictionary--> {'rice': 130, 'wheat': 175, 'maize': 60, 'pulse': 45}
crops--> dict_keys(['rice', 'wheat', 'maize', 'pulse'])
yield--> dict_values([130, 175, 60, 45])
 is sugarcane present--> False

 sum--> 420
```

In [44]:

```
for i in zip(crops.keys(),crops.values()):
  print(i)
crops2=crops
crops.copy()
print(crops)
print(crops2)
crops.clear()
print(crops)
```

```
('rice', 130)
('wheat', 185)
('maize', 60)
('pulse', 45)
{'rice': 130, 'wheat': 185, 'maize': 60, 'pulse': 45}
```

```
{'rice': 130, 'wheat': 185, 'maize': 60, 'pulse': 45}
{}
```

In [53]:

```
T=("anoop",'abhi',3.14,3)
print("T-->",T)
print("number of elements-->",len(T))
print("type of object-->",type(T))
```

```
T--> ('anoop', 'abhi', 3.14, 3)
number of elements--> 4
type of object--> <class 'tuple'>
```

In [62]:

```
T=("anoopG")
print("T-->",T)
print("number of elements-->",len(T))
print("type of object-->",type(T))
```

```
T--> anoopG
number of elements--> 6
type of object--> <class 'str'>
```

In [63]:

```
T=("anoop",'abhi',3.14,3)
print("T-->",T)
i=0
print("")
while i <len (T):
  print(T[i])
  i+=1
```

```
T--> ('anoop', 'abhi', 3.14, 3)

anoop
abhi
3.14
3
```

In [ ]:

```
```